

AI & ML Task 2 — Model Comparison Report

Feature Engineering, Model Optimization & Performance Comparison

Methodology

This task extends the Task 1 House Price Prediction project to a structured model-comparison workflow. The California Housing dataset (scikit-learn) was loaded with median house value as the target and the remaining eight numeric attributes as input features. Features were standardized using StandardScaler so that all inputs share a common scale, then split into training (80%) and test (20%) sets with a fixed random seed (random_state=42) for reproducibility.

Three regression models were trained and evaluated on the identical train/test split: Linear Regression (baseline), Ridge Regression with alpha=1.0 (L2-regularized, to reduce overfitting), and a Decision Tree Regressor with max_depth=5 (non-linear, to capture feature interactions the linear models cannot). Each model was scored on the held-out test set using RMSE (Root Mean Squared Error) and R^2 Score (proportion of variance explained).

Results

The table below reports test-set performance for all three models, computed directly from the notebook (AI_ML_Task2_Model_Comparison.ipynb):

Model	RMSE	R2 Score
Linear Regression	0.7456	0.5758
Ridge Regression (alpha=1.0)	0.7456	0.5758
Decision Tree (max_depth=5)	0.7242	0.5997

Linear Regression and Ridge Regression produced nearly identical results (RMSE 0.7456, R^2 0.5758), indicating that L2 regularization with alpha=1.0 had negligible effect on this dataset — consistent with the model not being strongly overfit to begin with. The Decision Tree Regressor achieved the lowest RMSE (0.7242) and highest R^2 Score (0.5997), explaining approximately 60% of the variance in house prices versus approximately 58% for the linear models, by capturing non-linear relationships between features such as median income, location, and house value.

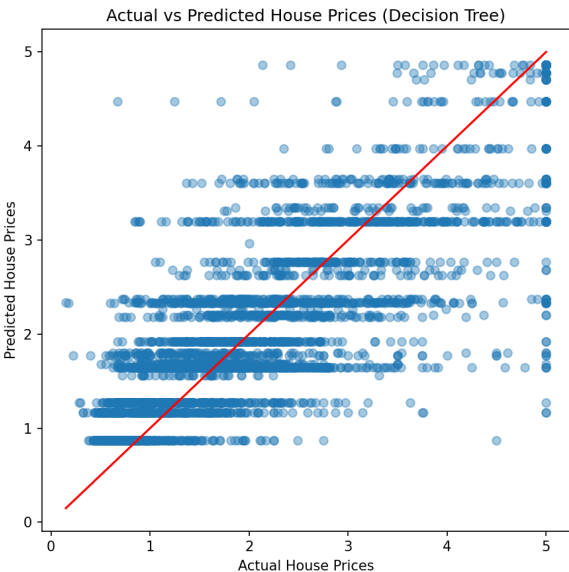


Figure — Actual vs Predicted House Prices for the selected best model (Decision Tree, max_depth=5). Points closer to the diagonal red reference line indicate more accurate predictions.

Conclusion

Based on the comparison above, the **Decision Tree Regressor (max_depth=5)** was selected as the final model, as it achieved the lowest RMSE and highest R^2 Score among the three candidates on unseen test data. This result illustrates a core lesson of the task: relationships between income, location, and house value in this dataset are not purely linear, so a model flexible enough to capture non-linear patterns and feature interactions outperforms both linear baselines. Ridge Regression's near-identical performance to plain Linear Regression suggests multicollinearity/overfitting was not a major issue for the linear approach here, so the gain from Ridge alone was minimal.

The selected model and its fitted StandardScaler were saved with joblib (task2_best_model.pkl, task2_scaler.pkl) for future reuse. A natural next step would be tuning the Decision Tree's depth (or trying ensemble methods such as Random Forest or Gradient Boosting) to test whether further gains are available beyond this single-tree baseline.